2nd Semester Topology Exam
January, 2026

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper (10 pts each problem).

1. Show that for every continuous map $f : S^1 \to S^1$ there is $n \in \mathbb{Z}$ such that f is homotopic to $z^n : S^1 \to S^1$.
2. Derive the Brouwer Fixed Point Theorem for B^2 from the Borsuk-Ulam Theorem for S^2.
3. Show that S^2 is not homeomorphic to S^3.
4. Show that every map $f : S^3 \to S^1 \times S^1 \times S^1$ of the 3-sphere to the 3-torus is null-homotopic.
5. Show that the Stone-Čech compactification of the reals is not metrizable.

Answer the following with complete definitions or statements or proofs (5 pts each problem).

6. Compute the fundamental group $\pi_1(S^2)$.
7. Show that any two continuous maps $f, g : X \to I^n$ to the n-cube are homotopic.
8. Is the space I^I separable where $I = [0, 1]$?
9. State the Seifert-van Kampen Theorem.
10. State the Tychonoff Theorem.
11. Show that the projective plane $\mathbb{R}P^2$ is not homeomorphic to the surface of genus 2, $M_2 = T\#T$.
12. Let $X = [0, 1]^{(0,1)}$ be given the product topology. Is the space X
(a) compact?
(b) metrizable?
(c) separable?
13. Let $X = \mathbb{R}$ and $A = \mathbb{Z}$ be the reals and the integers respectively.
(a) Is the quotient space $X^* = X/A$ metrizable?
(b) What is $\pi_1(X^*)$?
Here X^* is the partition into A and singletons.
14. Give definition of a deformation retraction.
15. If $r : X \to A$ is a retraction, show that the induced map for the fundamental groups $r_\#$ is surjective.